University Roll No._____

GLA UNIVERSITY, MATHURA
B. Tech. II Year (Even Semester), I Term Examination, 2015-16

Subject: English for Professional Purposes–II | Subject Code: AHE 4080
Time: 50 Minutes | Max. Marks: 15

Notes:

- Read the instructions carefully before you answer the question.
- Answer all questions from Section A, any 3 out of 4 from Section B, and any 2 out of 3 from Section C.
- Answers should be to-the-point, and word limit for informative questions from Section B (approx 60 words) and Section C (150 words) should strictly be maintained.
- All questions of a particular section should be answered at one place.
- Figures on the right-hand margin indicate marks.

Section A **0.5 x 6=3**

7. The order of academic qualification in a CV is arranged in _____ manner.
8. Extempore is a type of impromptu. (True/False)
9. Write the career objective of your CV.
10. Name any two occupations where characteristic norms of debating are considered to be a professional requirement.
11. Extempore and debate improves a person's _____ skill.
12. Write any two sentences/phrases that can be used for disagreeing in a GD.

Section B **2x3=6**

5. Why do companies hold GD, not Debate, for placement?
6. Write down the essential components of resume in specified order.
7. On what parameters can an extempore be judged? Explain.
8. An unsubstantiated argument in a debate is worthless. Comment.

Section C **3x2=6**

4. How is the initiation to a debate different from that of a GD?
5. Write five points each for proposition and opposition for any one of the given two:
   (c) Public health affects the economy of a nation.
   (d) Falling crude oil price is helping Indian economy.
6. You are applying for the post of GET at Alpha Engineering Solutions, Bangalore. Draft a cover letter. No CV/Résumé is required.

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University Roll No. | | | | | | | | | |

I Mid- Term Examination, 2015-16

Discrete Mathematics Paper Code - (CSE 4005)

Time: 90 Minutes Max. Marks: 20

Section – A

Note: Attempt All Questions. (1×5 = 05 Marks)

Q.1 If A and B are two sets such that n(A – B) = 24, n(B – A) = 19 and n(A ∩ B) = 11 find (i) n(A), (ii) n(B), (iii) n(A ∪ B)

Q.2 Find total number of relations from set A to set B if n (A) = 2, n (B) = 3.

Q.3 Find the number of permutations of the letters in the name TIMTIM.

Q.4 what is the minimum number of students required in a class to be that at least five will receive the same grade if there are four possible grades A, B, C, D.

Q.5 Find the generating function of the following discrete numeric functions 0, 1, -2, 4, -8,.....

Section – B

Note: Attempt Any Three Questions. (2×3 = 06 Marks)

Q.1 If A, B, C be three sets then prove that $A - (B \cup C) = (A - B) \cap (A - C)$

Q.2. Solve the recurrence relation $a_r - 3a_{r-1} + 2a_{r-2} = 2^r$

Q.3 How many permutations can be made out of the letter of word "COMPUTER"? How many of these:

(i) Begin with C, (ii) begin with C and end with R

(iii) End with R; (iv) C and R occupy the end places.

Q.4 Use Venn diagram to determine validity of following arguments;

P_1 : All my friends are musicians.

P_2 : John is my friend.

P_3 : None of my neighbours are musicians.

P : John is not my neighbour.

Section – C

Note: Attempt Any Three Questions. $(3 \times 3 = 09$ Marks)

Q.1 Among the first 500 positive integers

(a) Determine the integers which are not divisible by 2, nor by 3, nor by 5.

(b) Determine the integers which are exactly divisible by one of them.

Q.2 if R be a relation in the set $N \times N$ where N is a set of positive integers defined by $(a, b)R(c, d) \Leftrightarrow a + d = b + c, \forall a, b, c, d \in N$, then prove that R is an equivalence relation.

Q.3. Solve the recurrence relation $a_{r+2} - 2a_{r+1} + a_r = 2^r$ by method of generating functions with the initial condition $a_0 = 2$ and $a_1 = 1$.

Q.4 A box contains 5 different red and 6 different white balls, In how many ways can 6 ball be selected so that there are at least two balls of each type.

SET-A

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), Term I Examination, 2015-16

Subject: English for Professional Purposes–II
Time: 40 Minutes

Subject Code: AHE 4080
Max. Marks: 20

VERBAL APTITUDE

Spot the errors in the following sentences.

1. The result of the recognition (A)/ of these facts are seen (B)/ in the gradual improvement (C)/ of the diet of the poor (D)./ No Error (E).
2. Most of my relatives (A)/ give me an advice (B)/ when I am in problem (C)/ and I need somebody to help me (D). /No Error (E).
3. Taking bath in cold water in the morning (A)/ particularly in December (B)/ is one of the most horrible things (C)/ for me (D). /NO Error (E).

Sentences are given with blanks to be filled in with an appropriate word(s).Four alternatives are suggested for each question. Choose the correct alternative out of the four.

4. Every job _____its good points.
   A. have B. has C. are D. was E. is
5. Honour and glory _____his reward.
   A. is B. are C. have D. has E. were
6. The Mayor, with his councilors, _____to be present.
   A. is B. are C. have D. has been E. have been
7. Not only Jyoti but also her friends _____ been called to the principal's office.
   A. is B. are C. have D. has E. was
8. Rice and beans, my favorite dish, _____ me of my native Puerto Rico.
   A. reminds B. remind C. is reminded
   D. was reminded E. were reminded
9. Rajnish is full of zeal but unfortunately he has _____ skill.
   A. little B. some C. none D. few E. any
10. Ramesh is _____ Rahul Dravid of our team.
    A. a B. an C. the D. no article E. any
11. _____ money that I earned in the previous project has been invested in property.
    A. little B. A little C. the little D. few E. the few
12. Did you see _____ girls in the classroom when you entered?
    A. some B. any C. few D. most E. much

Identify the most preferable sentence in terms of grammar, meaning and usage in formal writing.

13. A. Our happiness or our sorrow are largely due to our own actions.
    B. Our happiness or our sorrow is largely due to our own actions.
    C. Our happiness or our sorrow have largely due to our own actions.
    D. Our happiness or our sorrow have been largely due to our own actions.
    E. None of the above.
14. A. Great news has been circulated by me.
    B. Great news have circulated by me.
    C. Great news has circulated by me.
    D. Great news circulated by me.
    E. None of the above.
15. A. Jane's raincoat, with all the rips and tears, provides little protection.
    B. Jane's raincoat, with all the rips and tears, provide little protection.
    C. Jane's raincoat, with all the rips and tears, has provide little protection.
    D. Jane's raincoat, with all the rips and tears, have provide little protection.
    E. None of the above.
16. A. Twenty kilometers is a long walk.
    B. Twenty kilometers are a long walk.
    C. Twenty kilometers have a long walk.
    D. Twenty kilometers make a long walk.
    E. None of the above.

Each of the following questions contains a small paragraph followed by a question on it. Read each paragraph carefully and answer the question given below it.

17. The virtue of art does not allow the work to be interfered with or immediately ruled by anything other than itself. It insists that it alone shall touch the work in order to bring it into being. Art requires that nothing shall attain the work except through art itself.
    This passage best supports the statement that:
    A. art is governed by external rules and conditions
    B. art is for the sake of art and life
    C. art is for the sake of art alone
    D. artist realizes his dreams through his artistic creation
    E. artist should use his art for the sake of society
18. Most social network services are web-based and provide means for users to interact over the Internet, such as e-mail and instant messaging. Online community services are sometimes considered as a social network service, though in a broader sense, social network service usually means an individual-centered service whereas online community services are group-centered. Social networking sites allow users to share ideas, pictures, posts, activities, events, and interests with people in their network.
    The passage best supports the statement that:
    A. Facebook service consists of a representation of each user (often a profile), his/her social links, and a variety of additional services.
    B. A social networking service is a platform to build social networks or social relations among people who, for example, share interests, activities, backgrounds, or real-life connections.
    C. Several websites are beginning to tap into the power of the social networking model for philanthropy.
    D. Companies have begun to merge business technologies and solutions, such as cloud computing, with social networking concepts.
    E. None of the above.

In each of the following questions, identify which part of speech is the underlined word.

19. There is nobody that can help me.
A. Adverb B. Pronoun C. Adjective D. Noun E. Conjunction
20. He is very rich still he is unhappy.
A. Adverb B. Conjunction C. Adjective D. Preposition E. Pronoun
21. This pen was gifted to me by my grandfather.
A. Adverb B. Preposition C. Pronoun D. Adjective E. Preposition
22. He left without paying his bill.
A. Adverb B. Conjunction C. Preposition D. Adjective E. Pronoun
23. Perhaps he is suffering from depression.
A. Adjective B. Adverb C. Preposition D. Conjunction E. Noun

Spot the errors in the following sentences.

24. She shifted her residence(A)/ to this city to be (B)/ close with the child (c)/she had waited to adopt (D)/ No error (E)
25. When the college(A)/ was established (B)/ he was (C) / yet practicing law (D)/No error(E)
26. I have (A)/ neither spoken to her (B)/ or written (C)/ to her (D)/No Error (E)
27. It will be difficult to(A)/ coordinate among the members(B)/ as each person who comes(C)/ will have their own views to present(D)/No error(E)
28. Our meticulous plan(A)/ came up to nothing (B)/ as he had to leave (C)/ for Delhi at a short notice(D)/ No Error (E)

Fill In The blanks with the proper form of the verbs given below.

29. Beto ____ in the shower generally.
a. Is singing
b. sing
c. has been singing
d. sings
e. has sung.
30. You ____half an hour in the network last night.
a. were spending
b. spent
c. have spent
d. will spent
e. have been spenting.
31. The children will be laughing while the clown ____ on one leg with a huge drum.
a. starts dancing
b. will be dancing
c. will have danced
d. would have danced
e. will have been dancing
32. She had sacked him before he ____ a chance to explain his behaviour.
a. had had
b. have had
c. would have
d. can have
e. had

Spot the errors in the following sentences.

33. I will call you (A) /and your friends (B) /with their classmates (C) /when the dinner will be ready (D)./ No error (E).
34. At this rate (A)/ she will be finished (B) /watching all Sherlock Holmes movies (C) /before we do (D)./ No error (E).
35. We need not to (A)/ understand (B)/ the information (C)/ as it was strictly on a need-to-know (D)./ No error (E).

Choose the correct alternative for the underlined parts.

36. In India today, many of our intellectuals still talk in terms of the French Revolution and the Rights of Man, not appreciating that much has happened since then.
A. much has been happening
B. much had happened
C. much might happen
D. much might have happened
E. No correction required
37. As the crops are dying, it must not had rained this much.
A. must not have had
B. must not have
C. must not have been
D. No correction required
38. If I was you, I would close the door and would eat to the last bit.
A. I were
B. I am
C. I would be
D. I was
E. No correction required

Read each paragraph carefully and answer the question given below it.

The relationship between my mother, sister and me had been cold and inimical for as long as I could remember. To me, my mother was irrational, hurling hurtful invectives for the slightest infraction. "Doesn't matter," my mother grumbled when I asked her where she moved my water-colour paints. "It ain't like you got talent." The time our mailbox got knocked off, she somehow got it in her head that I was the culprit. "Never respected nothing'," I heard her say. And my sister took my mother's side against me every time. Five years my senior, Tammy seemed not to have a brain in her head. She dithered about everything, incapable of making any firm decision. No matter how often my mother deprecated her — "dumb, ugly, fat" — Tammy made futile attempts to fawn her way back into mother's good graces.

39. Which is most likely to make a relationship inimical?
A. great pathos
B. frequent maledictions
C. magnanimous gestures
D. ingenious discussions
E. heated debates
40. If the story were true, which would best describe its genre?
A. literary essay, based on a piece of literature
B. memoir essay, centred on a significant memory from the past
C. persuasive essay, characterized by choosing a side and refuting other arguments
D. expository essay, meant to inform the reader of a body of knowledge
E. satire, in which human folly and vice are held up to scorn, derision, or ridicule

SET- C

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), Term I Examination, 2015-16

Subject: English for Professional Purposes–II
Time: 40 Minutes

Subject Code: AHE 4080
Max. Marks: 20

VERBAL APTITUDE

Spot the errors in the following sentences.

1. In January 1990,(A)/ the international news media was excited (B)/because it appeared (C)/that Mandela might be released (D)./ No Error (E).
2. Neither the coach came (A)/ nor did he let few players come (B)/ to participate (C)/in the tournament (D)./ No Error (E).
3. The love is the noblest quality (A)/ of human heart (B)/ and a person without love (C)/ is no better than a wild animal (D)./NO Error (E).

Sentences are given with blanks to be filled in with an appropriate word(s).Four alternatives are suggested for each question. Choose the correct alternative out of the four.

4. Each of the teachers _____very kind.
   A. was B. were C. have D. has
   E. none of the four
5. Bread and Butter _____only food for him.
   A. is B. are C. have D. has
   E. none of the four
6. The International Club, as well as the Choral Society and the Rowing Club, _____ to submit a new constitution.
   A. need B. needs C. have needed
   D. has needed E. none of the four
7. Each of the students in this building _____ for a new facility by the next year.
   A. has hoped B. hope C. hopes D. have hoped
   E. none of the four
8. A large number of voters still _____ along straight-party lines.
   A. vote B. votes C. is voted D. was voted
   E. none of the four
9. Pack your bag and leave now for the station. You hardly have _____ time for breakfast.
   A. few B. no C. any D. little E. a little
10. I have _____ European friend, who is a student at Oxford University.
    A. a B. an C. the D. no article E. none
11. Sorry, I cannot help you at this time. I have _____ money in my pocket.
    A. few B. a few C. the few D. little
    E. the little
12. I have not eaten _____ breakfast yet.
    A. some B. any C. few D. much E. none

Identify the most preferable sentence in terms of grammar, meaning and usage in formal writing.

13. A. Neither the Chairman nor the directors is present.
    B. Neither the Chairman nor the directors are present.
    C. Neither the Chairman nor the directors have present.
    D. Neither the Chairman nor the directors was present.
    E. None of the above.
14. A. The Committee has issued its report.
    B. The Committee have issued its report.
    C. The Committee has been issued its report.
    D. The Committee have been issued its report.
    E. The Committee has issued it's report.
15. A. Something in that curry make me sneeze.
    B. Something in that curry making me sneeze.
    C. Something in that curry makes me sneeze.
    D. Something in that curry maked me sneeze.
    E. None of the above.
16. A. Three parts of the property are left for me.
    B. Three parts of the property lefting for me.
    C. Three parts of the property is left for me.
    D. Three parts of the property are being left for me.
    E. None of the above.

Each of the following questions contains a small paragraph followed by a question on it. Read each paragraph carefully and answer the question given below it.

17. The argument 'irrespective of the economic progress we make, we continue to remain socially disadvantaged' by those benefitting from it is unacceptable. Reservation destroys self-respect, so much so that competition is no longer on to determine the best but the most backward. Is it fair to extend reservation to the children of professors, scientists, bureaucrats, MPs and MLAs belonging to the SC, the ST and the OBC?
    The passage best supports the statement that:
    A. Reservation in promotion is a natural and logical corollary of reservation at the entry level.
    B. The reservation system has received a mixed response from Indians since its inception.
    C. India never needs a reservation system.
    D. It was a mistake to differentiate people using reservation policy.
    E. None of the above.
18. One of the important humanitarian by-products of technology is the greater dignity and value that it imparts to human labour. In a highly industrialized society, there is no essential difference between Brahmin and Dalit, Muslim and Hindu; they are equally useful and hence equally valuable for, in the industrial society, individual productivity fixes the size of the pay cheque, and this fixes social status.
    The passage best supports the statement that:
    A. technology decides individual's social status.
    B. castes and religions are man-made.
    C. human labour has dignity and value.
    D. all individuals, irrespective of caste and creed, are born equal.
    E. industrial society is a great leveller of men.

In each of the following questions, identify which part of speech is the underlined word.

19. What is <u>that</u> troubles you.

A. Pronoun B. Adjective c. Adverb. D. conjunction E. Preposition

20. He must be very hungry <u>for</u> he has not eaten anything.
    A. Preposition B. Conjunction C. Adjective D. Adverb E. Pronoun
21. <u>That</u> book is not in the reference section of the library.
    A. Adverb B. Adjective C. Pronoun D. Preposition E. Conjunction
22. He went to Australia <u>without</u> her.
    A. Preposition B. Adjective C. Adverb D. Conjunction E. Pronoun
23. <u>Perhaps</u> he has forgotten his book in the classroom.
    A. Adjective B. Adverb C. Conjunction D. Preposition E. Pronoun

Spot the errors in the following sentences.

24. As she is getting old (A)/ her understanding of (B)/ human frailties is (C)/ becoming deeper (D)./ No Error (E).
25. I would swear on a thousand Gods (A)/that my friend Vikram almost (B)/drowned once when (C)/he was visiting his aunt (D)./No error (E).
26. The man told to her that (A)/ he had not brought his dog out (B)/for a walk as he was afraid(C)/ that it would rain(D)./No error (E).
27. The Union Health Minister said that (A)/there was an acute shortage of health personnel (B)/in rural areas who needed (C)/to be addressed urgently (D)./No Error (E).
28. As I was to reach (A) / early I left in aeroplane (B) / instead of (C) / going by train (D). / No error (E).

Fill In The blanks with the proper form of the verbs given below.

29. Don't call me until about 7.30. I _____my homework until then.
    A. will not have been finishing
    B. won't have finished
    C. won't finish
    D. will not be finishing
    E. cannot finish
30. She asked me if I wanted to go to the cinema, but I _____in the garden the whole day and all I wanted to do was go to bed.
    A. was working
    B. had worked
    C. had been working
    D. worked
    E. am working
31. At the end of the next month, we _____ for exactly 25 years.
    A. Will have been married
    B. Will marry
    C. Will be married
    D. will have married
    E. would be married
32. She doesn't realise what kind of person he is, but she _____ .
    A. will be finding out
    B. will find out
    C. will have found out
    D. will found out
    E. would find out

Spot the errors in the following sentences.

33. The principal announced (A)/ in the school assembly that the school fees (B) /will be hiked (C) / from the beginning of the next academic session (D). / No error (E).
34. They have recently began (A)/ to purchase (B)/ furniture for their new house (C)/ which is being constructed in Poona (D). / No error (E).
35. Rehman says that (A)/ Ramesh takes charge of (B)/ the project within a few days (C)/ of appointment (D)./ No error (E).

Choose the correct alternative for the underlined parts.

36. A: 'You look tired.'
    B: 'Yes, <u>I've played</u> basketball since this afternoon.'
    A. I've been playing
    B. I was playing
    C. I played
    D. I play
    E. No correction.
37. I wonder why Jim <u>is</u> so nice to me today. He isn't usually like that.
    A. is being
    B. has been
    C. is behaving
    D. behaves
    E. No correction.
38. <u>Does he live</u> in London this month?
    A. Is he living
    B. Is he currently living
    C. Has he been currently living
    D. was he living
    E. No correction.

Read the passage carefully and answer the questions given below.

Jeffrey brushed quickly past an elderly woman waiting on the platform ahead of him to get onto the subway. He wanted to be sure to get a seat to read his *New Economics Journal*. As the train screeched out of the station, he lifted his head from the business news and stared at the man directly across him. A fierce wave – a tsunami – of antipathy came over him. Jeffrey knew this man, knew him all too well. They had become bitter enemies. Their eyes locked.

As the train reached full speed, the cacophony of speeding wheels against the winding rails and of the wildly vibrating subway car filled Jeffrey's ears. To this frenetic beat, Jeffrey listed in his head all the reasons that this man, whose eyes he gravely stared into, whose image seemed to appear nearly everywhere he went, had become an anathema to him.

39. "A fierce wave – a tsunami – of antipathy came over him." Given the imagery in this sentence, which word would be the best substitute for came?
    A. Angled
    B. Drizzled
    C. Rained
    D. Washed
    E. trampled
40. As used here, anathema is best defined as a(n):
    A. Curse
    B. Vision
    C. Embarrassment
    D. Antagonist
    E. problem

SET-B

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), Term I Examination, 2015-16

Subject: English for Professional Purposes-II
Time: 40 Minutes

Subject Code: AHE 4080
Max. Marks: 20

VERBAL APTITUDE

Spot the errors in the following sentences.

1. Cows are amongst (A)/ the gentlest of breathing creatures ; (B)/ none other shows more passionate tenderness (C)/to their young (D)./ No Error (E).
2. The trekkers decided to return to their base camp (A)/which was at few distance (B)/from where they were (C)/at that point in time (D)./ No Error (E).
3. Decision taken by (A)/ the members was rejected (B)/ though initially it had been (C)/accepted by the authorities (D)./ No Error (E).

Sentences are given with blanks to be filled in with an appropriate word(s).Four alternatives are suggested for each question. Choose the correct alternative out of the four.

4. The homework for those classes _____ very time-consuming.
   A. is B. are C. have D. has E. none of the four
5. My friend and benefactor _____ come.
   A. is B. are C. have D. has E. none of the four
6. One of my best friends _____ qualified the UPSC exam this year.
   A. is B. are C. have D. has E. none of the four
7. Each of the students_____for a new facility by the next year.
   A. has hoped B. hope C. hopes D. have hoped E. none of the four
8. Four years _____ a long time to spend away from your friends and family.
   A. is B. are C. have D. has E. none of the four
9. I am comfortable in this town as I have _____ acquaintances here.
   A. few B. a few C. the few D. little E. the little
10. The professor of Mathematics and _____ Dean of Research and Development is coming.
    A. a B. an C. the D. no article E. some
11. If you do not leave now, you will miss the train. You do not have _____ time to waste.
    A. little B. any C. many D. few E. some
12. Are there _____ bananas in the refrigerator?
    A. some B. any C. few D. much E. none

Identify the most preferable sentence in terms of grammar, meaning and usage in formal writing.

13. A. He asked me whether either of the applicants was suitable.
    B. He asked me whether either of the applicants were suitable.
    C. He asked me whether either of the applicants has suitable.
    D. He asked me whether either of the applicants was being suitable.
    E. None of the above.
14. A. The Committee is divided on one point.
    B. The Committee are divided on one point.
    C. The Committee is being divided on one point.
    D. The Committee was divided on one point.
    E. None of the above.
15. A. The ebb and flow of the tides is explained by Newton.
    B. The ebb and flow of the tides are explained by Newton.
    C. The ebb and flow of the tides explained by Newton.
    D. The ebb and flow of the tides is explaining by Newton.
    E. None of the above.
16. A. Two third of the city is in ruins.
    B. Two third of the city are in ruins.
    C. Two third of the city being in ruins.
    D. Two third of the city be in ruins.
    E. None of the above.

Each of the following questions contains a small paragraph followed by a question on it. Read each paragraph carefully and answer the question given below it.

17. The attainment of individual and organizational goals is mutually interdependent and linked by a common denominator - employee work motivation. Organizational members are motivated to satisfy their personal goals, and they contribute their efforts to the attainment of organizational objectives as means of achieving these personal goals.
    The passage best supports the statement that motivation –
    A. encourages an individual to give priority to personal goals over organizational goals.
    B. is crucial for the survival of an individual and an organization.
    C. is the product of an individual's physical and mental energy.
    D. is the external force which induces an individual to contribute his/her efforts.
    E. makes organization and society inseparable.
18. Due to enormous profits involved in smuggling, hundreds of persons have been attracted towards this anti-national activity. Some of them became millionaires overnight. India has a vast coastline both on the Eastern and Western Coast. It has been a heaven for smugglers who have been carrying on their activities with great impunity. There is no doubt that from time to time certain seizures were made by the enforcement authorities during raids and ambushes but even after these losses the smugglers made huge profits.
    The passage best supports the statement that
    A. smuggling hampers the economic development of a nation.
    B. smuggling ought to be curbed.
    C. authorities are taking strict measures to curb smuggling.

D. smuggling is fast increasing in our country owing to the quick profit it entails.
E. None of the above

In each of the following questions, identify which part of speech is the underlined word.

19. I don't say anything that can hurt others.
A. Adjective B. Adverb C. Pronoun D. Noun E. Preposition
20. I was upset still I kept quiet.
A. Adverb B. Conjunction c. Preposition D. Adjective E. Pronoun
21. This house is mine.
A. Noun B. Pronoun C. Adjective D. Adverb E. Conjunction
22. I do not want to go without you.
A. Preposition B. Adverb C. Conjunction D. Adjective E. Conjunction
23. Perhaps you are right.
A. Conjunction B. Adverb C. Preposition D. Adjective E. Pronoun

Spot the errors in the following sentences.

24. Because of his ill health(A)/ the doctor has advised(B)/ him to (C)/ refrain to smoking(D)./No error (E).
25. The man who(A)/ has committed such a serious crime(B)/ must get the (C)/mostly severe punishment (D)./No Error (E).
26. We will pack not only (A)/ the material properly (B)/ but will also deliver it (C)/ to your valued customers (D)./ No error (E).
27. If you are living (A)/near a market place (B)you should be ready to bear the(C)/ disturbances caused by traffic (D)./No Error (E).
28. If you are not clear (A)/ about the meaning (B)/of a word, it is wise (C)/ to look to a dictionary (D)./ No Error (E).

Fill in the blanks with the proper form of the verbs given below.

29. If she_____then I will provide accommodation.
A. would be ready staying
B. is ready to stay
C. be ready staying
D. will stay
E. need staying
30. Before the teacher came to class, the students_____ their assignments.
A. were completing
B. had completed
C. had had completed
D. would complete
E. completed
31. I arrived over an hour late to the office and everyone was working. Actually, they _____for over two hours on the new project and I felt really guilty.
A. had worked
B. have worked
C. were working
D. worked
E. could work
32. They would have won if they _____ better.
A. had played
B. will play
C. have played
D. have play
E. could play.

Spot the errors in the following sentences.

33. He started a very small business two years ago (A)/ but it grew very fast (B)/ as the country is experiencing(C)/ a boom at that time (D)./ No error (E).
34. Had I known (A)/ the weather forecast earlier (B)/ I would not (C)/ make plans to go out (D)./ No error (E).
35. Big landlords in the villages(A) /who were till now paid for electricity(B)/at subsidized rates will soon (C)/ have to pay the government at full rates (D)./ No error (E).

Choose the correct alternative for the underlined parts.

36. Everything is going well. We haven't have any problems so far.
A. haven't had
B. did have
C. could not have
D. had had
E. No corrections.
37. I wasn't very busy. I wasn't having much to do.
A. didn't have
B. hadn't have
C. wasn't having.
D. had had nothing
E. No correction.
38. A: 'Ann has gone out.'
B: 'Oh, has she? What time did she go?'
A. has she gone
B. did she left.
C. were she gone.
D. had she gone
E. No correction.

Read the passage carefully and answer the questions given below.

"Nolli!" Mother Ann shouted suddenly, her thunderous voice tantamount to the rumbling storm Rowan was attempting to capture in her reading. Nolan gave a little start. "You have no idea how difficult it is to speak in front of a crowd. Your sister has been preparing for this presentation for weeks and I swear if you don't..."

Mother Ann continued to castigate Nolan most thoroughly, scolding him for his rudeness and lack of respect. Then, feeling as though she had achieved the intended effect, she sighed rather contentedly and reclined in her chair. Sometimes she reminded even herself the fury by which, if completely necessary, she could quell any quarrel.

Nolan had fallen silent and was now sitting upright, his hands folded before him.

39. If X is tantamount to Y, then X
A. is longer than Y
B. is less than Y
C. equals Y
D. and Y have no connection
E. is above Y
40. If Mother Ann is castigating Nolan, what is she doing?
A. loudly yelling at him
B. mildly abusing him
C. vehemently threatening him
D. severely punishing him
E. coldly antagonizing him

First Term Examination, 2015-16

Course: B.Tech, Year- II: , Semester:IV

Object Oriented Programming (CSE4004)

Uni. Roll No:

Time: **90** minutes Max. Marks: **20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section-A

Note: Attempt All Questions **1*5=5 marks**

1. Differentiate between class variable and instance variable?
2. Enlist any four UML diagrams?
3. Identify the relationship(association, aggregation, generalization or dependency) in the following statements
   a) Database consists of records b) A document contains many paragraph
4. Write the statement required to invoke y in the below code:

```
class Btech
{ int x=20;
  class CSE
{
int y=100;
} }
class OOPS
{
public static void main (String error[])
{
// Write statement here to invoke value of y
```

}}

5. Enlist any four packages defined in Java Standard library.

Section: B

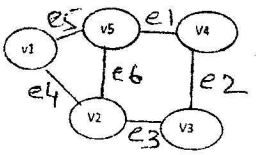
Note: Attempt Any Three Questions **2*3=6 marks**

1. Discuss the significance of static block in java? Enlist the rule for static method along with the use of static class in java?
2. Is it possible to use static data inside non static method? Justify your answer with an example?
3. Discuss briefly UML modeling types?
4. Design an instance diagram for below statement:
   A line intersects two points

Section:C

Note: Attempt Any Three Questions **3*3=9 marks**

1. Describe the below relationship along with their symbolic representation
   a) Dependency b) Aggregation c) Generalization
2. Discuss with example how polymorphism is implemented in java?
3. Prepare an instance diagram for this graph



4. Define constructor? Differentiate between constructor and method in java? Enlist different access modifiers used in java?

University Roll No. | | | | | | | | | |

I Mid- Term Examination, 2015-16

Discrete Mathematics — Paper Code - (CSE 4005)

Time: 90 Minutes — Max. Marks: 20

Section – A

Note: Attempt All Questions. (1 × 5 = 05 Marks)

Q.1 If A and B are two sets such that $n(A - B) = 24$, $n(B - A) = 19$ and $n(A \cap B) = 11$ find (i) $n(A)$, (ii) $n(B)$, (iii) $n(A \cup B)$

Q.2 Find total number of relations from set A to set B if $n(A) = 2$, $n(B) = 3$.

Q.3 Find the number of permutations of the letters in the name TIMTIM.

Q.4 what is the minimum number of students required in a class to be that at least five will receive the same grade if there are four possible grades A, B, C, D.

Q.5 Find the generating function of the following discrete numeric functions 0, 1, -2, 4, -8,.....

Section – B

Note: Attempt Any Three Questions. (2 × 3 = 06 Marks)

Q.1 If A, B, C be three sets then prove that $A - (B \cup C) = (A - B) \cap (A - C)$

Q.2. Solve the recurrence relation $a_r - 3a_{r-1} + 2a_{r-2} = 2^r$

Q.3 How many permutations can be made out of the letter of word "COMPUTER"? How many of these:

(i) Begin with C, (ii) begin with C and end with R

(iii) End with R; (iv) C and R occupy the end places.

Q.4 Use Venn diagram to determine validity of following arguments;

P_1 : All my friends are musicians.

P_2 : John is my friend.

P_3 : None of my neighbours are musicians.

P : John is not my neighbour.

Section – C

Note: Attempt Any Three Questions. $(3 \times 3 = 09$ Marks)

Q.1 Among the first 500 positive integers

(a) Determine the integers which are not divisible by 2, nor by 3, nor by 5.

(b) Determine the integers which are exactly divisible by one of them.

Q.2 if R be a relation in the set $N \times N$ where N is a set of positive integers defined by $(a, b)R(c, d) \Leftrightarrow a + d = b + c, \forall a, b, c, d \in N$, then prove that R is an equivalence relation.

Q.3. Solve the recurrence relation $a_{r+2} - 2a_{r+1} + a_r = 2^r$ by method of generating functions with the initial condition $a_0 = 2$ and $a_1 = 1$.

Q.4 A box contains 5 different red and 6 different white balls, In how many ways can 6 ball be selected so that there are at least two balls of each type.

FIRST Term Examination, 2015-16
B.Tech. CSE II Year IV Semester

Operating Systems (CSE 4001)

Uni. Roll No:.....

Time: 90 Min. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Write the fundamental difference between asymmetric and symmetric multiprocessing.

II. List the four major categories of the benefits of multithreaded programming.

III. Enlist the various CPU scheduling criterias.

IV. List out the various system calls for process management.

V. What do you understand by Context Switching?

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Write the essential properties of following systems:
   a. Time sharing systems b. Real time systems

II. Explain layered structure of an operating system. Also explain advantages and disadvantages of the layered approach to system design.

III. Differentiate between kernel-level and user-level threads.

IV. Write short notes on:
   a. Dual Mode Operation b. Memory protection

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Discuss various operating systems components.

II. Draw the process state transition diagram. Also define various types of queues with the help of queuing diagram and different schedulers used for execution of the processes.

III. Explain different multiple queue scheduling algorithms.

IV. Consider the set of processes given in the table and draw the Gantt chart for preemptive cases and find out the average waiting time and average turnaround time for following scheduling algorithms (*Larger priority number has higher priority)

a. Round Robin (q = 3) **b.** SRTF **c.** Priority

Process ID	Arrival Time	Execution Time	Priority
P0	0	5	4
P1	2	4	2
P2	2	2	6
P3	4	4	3

FIRST TERM EXAMINATION 2015-16

University Roll No.....

B. Tech. II Year IV Semester

CSE-4003: Computer Networks

Time: 01 ½ Hours **Total Marks: 20**

Note: Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.

Section-A

Note: Attempt all Questions **(1X5=5)**

Q1.In a digital transmission, the sender clock is 0.3 percent faster than the receiver clock. How many extra bits does the sender send if the data rate is 2 Mbps?

Q2.What are the values of SNR and SNR_{dB} for a noiseless channel?

Q3. Show the Manchester coding for data stream 01110110.

Q4. What do you mean by flow control and Access Control mechanism?

Q5. Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth.

Section-B

Note: Attempt any three questions **(2x3=6)**

Q1. Calculate the Shannon Capacity for extremely noisy channel. Suppose a signal travels through a transmission medium from point P1 to point P2 and its voltage is reduced to one half. Calculate the attenuation in dB.

Q2. Explain the relationship between the bit rate and baud rate in the analog transmission of digital data. An analog signal has a bit rate of 4000 bps and a baud rate of 500 baud. How many data elements are carried by each signal element? How may signal elements do we need?

Q3.What are the responsibilities of Data Link layer and Network layer in OSI Model.

Q4.A slotted ALOHA network transmits 400 bit frames using a shared channel with a 400 kbps bandwidth. The network contains 100 stations. What is the number of frames/sec each station can send to achieve maximum efficiency? Also explain the vulnerable period in slotted ALOHA.

Section-C

Note: Attempt any three questions **(3x3=9)**

Q1. Discuss the properties of chip sequences in CDMA. Find the chips for a network with 4 stations and 8 stations.

Q2. Explain the total delay in a circuit switched network. Suppose a path in circuit-switched network has a data rate of 2 Mbps. The exchange of 2000 bits is required for the setup and teardown phases. The distance between two parties is 4000 km and the propagation speed is 2×10^8 m/s. What is the total delay if 2000 bits of data are exchanged during the data transfer phase?

Q3. What do we mean when we say that a bridge can filter traffic? Let's consider the operation of a learning switch in the context of a network in which 6 nodes labeled A through F are star connected into an Ethernet switch. Suppose that (i) B sends a frame to E, (ii) E replies with a frame to B, (iii) A sends a frame to B, (iv) B replies with a frame to A. The switch table is initially empty. Show the state of the switch table before and after each of these events. For each of these events, identify the link(s) on which the transmitted frame will be forwarded, and briefly justify your answers.

Q4. Discuss Guided Transmission media in detail.

University Roll. No.

B.Tech II[nd] Year CSE (4[th] Semester)

1[st] Mid-Term Examination, 2015-16

Introduction to Microprocessor Paper Code: - **CSE-4002**

Time: - **90 Min.** Max Marks: - **20**

SECTION-A

Attempt ALL Questions. **(1 x 5 = 5)**

1. Define three byte instruction with one example.
2. What is the data bus size and memory capacity used for 8086 microprocessor?
3. Discuss (a) CPI data and (b) CMC instruction in 8085 microprocessor.
4. What is the difference between microprocessor and microcontroller?
5. What is the use of status signals (S_1, S_0) in 8085 microprocessor?

SECTION –B

Attempt ANY THREE questions. **(2 x 3 = 6)**

1. Draw the timing diagram for op-code fetch operation and Explain.
2. Explain the following instructions. Also indicate the flags condition.
   (a) RLC (b) RRC (c) RAL (d) RAR
3. What are the differences between CALL and JUMP instructions? Explain with suitable examples.
4. The contents of register (B) and Accumulator (A) of 8085 microprocessor are 49H and 3AH respectively. What will be the contents of A and the status of carry flag (CY) and sign flag (S) after executing SUB B instruction?

SECTION –C

Attempt ANY THREE questions. **(3 x 3 = 9)**

1. Draw the architecture of 8085 microprocessor and Explain.
2. Explain the following instructions giving suitable examples.
   (a) SHLD (b) XCHG (c) LDAX (d) CMA (e) STAX (f) DAD rp
3. Explain all the types of Addressing Modes in microprocessor. Give suitable examples.
4. Write an assembly language program of 16 bit addition with carry (16 bit numbers available in BC and DE register pair; transfer the result in HL register pair; then store the result and Carry in different memory locations).

First Term Examination,
Even Semester 2015-16

Course: B.Tech II Yr. **Univ. No.** ----------
Title of Paper: Environmental Studies
Paper Code: AHS-3001
Time: 1.5 Hours **Max. Marks**: 20

Section-A

Note: Attempt All Questions **1 X 5 = 5 Marks**

(I) Why astronaut does not feel high temperature in ionosphere?
(II) Differentiate synecology and autecology with example.
(III) Define xerarch.
(IV) What is food web?
(V) Write down composition of four major gases present in atmosphere.

Section-B

Note: Attempt Any three Questions **2 x 3 = 6 Marks**

(I) Write down major causes of crisis of potable water.
(II) Discuss the exploitation of mineral resources.
(III) Write down four main causes of deforestation.
(IV) Why is environmental study multi disciplinary in nature?

Section-C

Note: Attempt Any Three Questions **3 x 3 = 9 Marks**

(I) What is ecological pyramid? Explain pyramid of number and biomass with diagrams.
(II) Write note on lithosphere, hydrosphere and biosphere.
(III) Discuss problems due to excess intake of fluoride via drinking water and its remedial measures.
(IV) What do you mean by natural resources? How can you classify them and also write two examples of each?